

An Arbitrary Style Transfer Network based on Dual Attention Module

Yueming Wang

Faculty of Information Technology, Beijing University of Technology, Beijing, China
 stevenberg@sina.com

Abstract—Arbitrary style transfer means that stylized images can be generated from a set of arbitrary input image pairs of content images and style images. Recent arbitrary style transfer algorithms lead to distortion of content or incompleteness of style transfer because network need to make a balance between the content structure and style. In this paper, we introduce a dual attention network based on style attention and channel attention, which can flexibly transfer local styles, pay more attention to content structure, keep content structure intact and reduce unnecessary style transfer. Experimental results show that the network can synthesize high quality stylized images while maintaining real-time performance.

Keywords—neural network; arbitrary style transfer; dual attention; style transfer

I. INTRODUCTION

Image stylization transfer refers to making computer generate artistic image through computer algorithm. Image style migration refers to that the computer imitates a certain artistic style to generate artistic image of this style through computer algorithm, and extracts the overall artistic style of one image and displays the content of another image in this style. Before the methods of deep learning is proposed, there are many methods for image based artistic rendering (IB-AR) image style migration defined by Kyprianidis et al. [1], including stroke-based rendering (SBR), region-based rendering (RBR), example-based rendering (EBR) and image processing and filtering technology. These technologies can only work for a specific style, not reusable, and each style requires a lot of work. Recently, Gatys et al. [2] carried out pioneering work. They found that deep convolution neural network can extract feature information from any photo, which can be used to extract style information from any art image. According to this finding, Gatys et al. reconstruct the content information by reducing the difference of high-level features between the content image and the target image, and reconstruct the style information by reducing the difference in the statistical features of the Gram matrix between the style image and the target image. The results show that the correlation between the features extracted from the pre trained deep neural network can capture the style pattern and generate stylized image of the target style. The method of Gatys et

al. is flexible enough to combine the content and style of any image pair. However, due to the iterative optimization process, the image generation speed is very slow, and this leads to a degree of image semantic loss and image distortion.

In order to make the image style transfer obtain wider application scenarios, many scholars have made further improvement and optimization based on the research of Gatys et al. The main improvement direction comes from the following three points: the generation speed of stylized image, the migration model for multi styles, and higher quality of stylized image. In many applications, the generation speed of stylized images is an important indicator for applications. Therefore, the method of fast style transfer has been proposed rapidly.

The simplest way is to train a specific forward model for each style, and the model can convert the image of the specified style by only input a content map. The paper [5]-[6] adopts this idea, uses a forward network to learn a style, and uses Gram matrix to model the style. The results show that the pre trained model can quickly generate stylized images, but a model only corresponds to one style. If a new style image needs to be generated, the new model needs to be trained again. Subsequently, Ulyanov et al. [7] found that the convergence rate of the model can be greatly improved by replacing batch normalization (BN) with instance normalization (IN).

In order to solve this problem, many researchers began to study models that can learn multiple styles. Dumoulin et al. [8] found that on the basis of a trained stylized network, a stylized image with completely different styles can be obtained by making an affine transformation on the instance normalization layer, which they call conditional instance normalization (CIN). Experimental results show that only a small number of parameters of CIN layer are trained to learn the new style and keep the other parts unchanged, the image with new style can be generated. The similar idea is the method proposed by Chen et al. [9]. Their model includes three parts: Encoder E, StyleBank Layer K and Decoder D. E and D are used to building image content, and K is used to building image style. By separating content and style, when encountering a new style, a new K is initialized for training and holding E and D unchanged.

Further, researchers hope to use only one model, when encountering a new style, unlike the previous model, which needs training for the new style, the new model can complete the stylization of the new style without training. Chen [10] proposed fast patch-based style transfer of arbitrary style, but it does not achieve real-time stylization because the exchange of patch needs time. Huang [11] proposed an adaptive instance normalization (AdaIN), which aligns the input content with the mean and variance of the style features, and transfer the statistical information of features to make the real-time stylization of arbitrary style.

Park et al. [12] observed that the image quality generated by AdaIN is not sufficient. Inspired by [13]-[15], a novel style attention network SANet is proposed. Style attention network is used to replace adaptive instance normalization to complete style migration. Experiments show that style migration can be better. In addition, compared with the single model, the quality of image generated by SANet can be improved significantly.

Inspired by the reference [16], this paper makes improvement based on [12] and proposes a dual attention arbitrary style transfer network. Style transfer network is composed of encoder module and dual attention module. Dual attention module includes style attention module and channel attention module. Style attention module is used to pay attention to the semantic association between content features and style features, and channel attention is used to enhance the association ability of different semantics under channels.

II. METHODS

In this section, we propose a style transfer network with dual attention structure. The network can effectively

reflect the image style, maintain the image structure, and generate high quality stylized images. We first introduce the basic structure of the network, then introduce dual attention modules, and finally describe the loss function of the network.

A. Network Architecture

The overall architecture of our style transfer network is shown in the Fig. 1. The input to the network is a content image I_c and a style image I_s , generating stylized images I_{cs} . The network structure is mainly divided into three parts: encoder, dual attention module and decoder. We use a pre-trained VGG-19 [17] network as encoder and symmetric decoder. Two parallel dual attention modules are used for joint training to achieve style transfer. In order to combine the local style and the global style better, we use two groups of parallel dual attention modules to input different layers (Relu_4_1 and Relu_5_1) and then integrate two output feature maps.

Firstly, we use VGG to encode content image and style image $F_c = E(I_c)$ and $F_s = E(I_s)$, and extract the feature map from different layers F_c^{r-4-1} , F_s^{r-4-1} , F_c^{r-5-1} , F_s^{r-5-1} . Then the content feature map and style feature map from the same layer are sent to one dual attention module to generate the feature map which processed by the dual attention module F_d :

$$F_d = DA(F_c, F_s) \quad (1)$$

Next, combine the feature maps from two dual attention module. We combine F_d^{r-4-1} and F_d^{r-5-1} after a upsampling and then feed two feature map into a 3×3

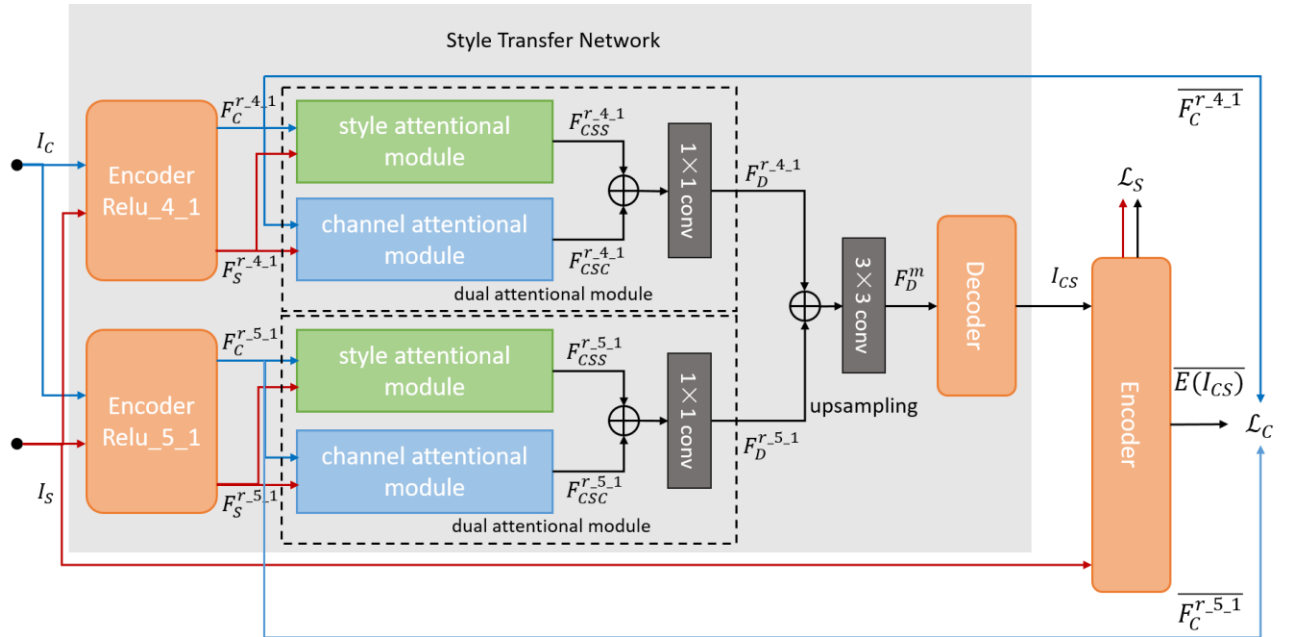
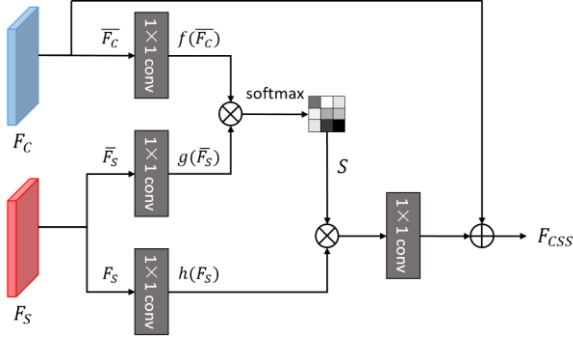


Fig. 1. Overview of style transfer network architecture



convolution layer. Then, we obtain final feature map F_D^m :

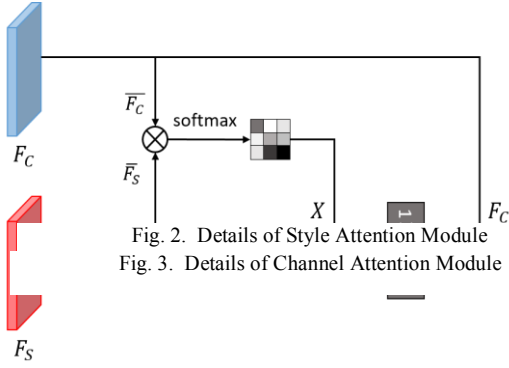


Fig. 2. Details of Style Attention Module
Fig. 3. Details of Channel Attention Module

$$F_D^m = \text{conv}_{3 \times 3} \left(F_D^{r-4-1} \oplus \text{upsampling} \left(F_D^{r-5-1} \right) \right) \quad (2)$$

where “ $\oplus$ ” denotes element-wise sum.

Finally, feed F_D^m into decoder to synthesize stylized output image I_{CS} .

B. Dual attention module

Style attention module. By learning the mapping relationship between the content feature map and the style feature map, the style attention module can embed the local style appropriately in each position of the content feature map. Fig. 2 shows the structure of the style attention module in detail.

As shown in the Fig. 2, a set of content feature F_C and style feature F_S as the input. We apply normalization to F_C and F_S and feed them into convolution layers to get two new feature maps $f(\overline{F_C})$ and $g(\overline{F_S})$, which shape are $C \times H \times W$, reshape them to $C \times N$, where $N = H \times W$. After that we perform a matrix multiplication between $f(\overline{F_C})$ and $g(\overline{F_S})$, and calculate the style attention map S by apply a softmax layer:

$$s_{ji} = \frac{\exp \left(f(\overline{F_C})_i g(\overline{F_S})_j \right)}{\sum_{i=1}^N \exp \left(f(\overline{F_C})_i g(\overline{F_S})_j \right)} \quad (3)$$

where s_{ji} represents the interrelation between the i^{th} position and j^{th} position. The more similar the feature representation of two styles, the greater the interrelation between them.

At the same time, we feed style feature F_S into a convolution layer to generate a new feature map $h(F_S)$. After a reshape, perform a matrix multiplication with transpose of S , and then reshape the result to make a final application of apply a element-wise sum with F_C to get F_{CSS} :

$$F_{CSSj} = \text{conv}_{1 \times 1} \left(\sum_{i=1}^N (s_{ji} h(F_S)_i) \right) + F_{Cj} \quad (4)$$

where the convolutions described above are learning weight matrices and are implemented as 1×1 convolutions.

Channel attention module. The channel map of feature can be regarded as the representation of different semantics. By establishing the correlation of channel maps, the interdependent features can be emphasized and the feature representation of specific semantics can be improved. Therefore, we propose to use the channel attention module to enhance the feature representation of the network.

Fig. 3 shows the structure of the channel attention module in detail. The module having a set of input content feature F_C and style feature F_S . But different from style attention module, we calculate channel attention map X only apply normalization to them, we get $\overline{F_C}$ and $\overline{F_S}$. Then, reshape them to $C \times N$, where $N = H \times W$, and perform a matrix multiplication between $\overline{F_C}$ and transpose of $\overline{F_S}$. At last, apply softmax to get channel attention map X :

$$x_{ji} = \frac{\exp \left(\overline{F_{C_i}} \overline{F_{S_j}} \right)}{\sum_{i=1}^C \exp \left(\overline{F_{C_i}} \overline{F_{S_j}} \right)} \quad (5)$$

where x_{ji} represents the interrelation between the i^{th} channel and the j^{th} channel. The more similar the feature representation of two channels, the greater the interrelation between them.

Next, we perform matrix multiplication between X and transpose of F_S , and reshape the result to $C \times H \times W$.

Then, feed in a convolution layer and apply a element-wise sum with F_c to get result F_{CSC} :

$$F_{CSC_j} = conv_{1 \times 1} \left(\sum_{i=1}^c (x_{ji} F_{Si}) \right) + F_{C_j} \quad (6)$$

where, similarly, the convolutions described above are learning weight matrices and are implemented as 1×1 convolutions.

Integration of two attention modules. Finally, we integrate the features of the two attention modules. The output of the two modules is transformed by a convolution layer, Integrate the feature by apply an element-wise sum to the outputs of two modules, and then transformed by a convolution layer to eliminate the aliasing effect:

$$F_D = conv_{1 \times 1} (F_{CSC} \oplus F_{CSS}) \quad (7)$$

C. Loss function

We use the pre-trained VGG-19 [17] as the encoder to calculate the loss function for training the network. The total loss can be defined as:

$$\mathcal{L} = \lambda_c \mathcal{L}_c + \lambda_s \mathcal{L}_s + \mathcal{L}_{identity} \quad (8)$$

where, total loss is consists of three part, content loss $\mathcal{L}_c$, style loss $\mathcal{L}_s$ and identity loss $\mathcal{L}_{identity}$. $\mathcal{L}_c$ and $\mathcal{L}_s$ have different weight.

The content loss is the Euclidean distance between normalized features $\overline{F_C^{r-4-1}}$, $\overline{F_C^{r-5-1}}$ and normalized features of output image $\overline{E(I_{CS})^{r-4-1}}$, $\overline{E(I_{CS})^{r-5-1}}$:

$$\mathcal{L}_c = \left\| \overline{E(I_{CS})^{r-4-1}} - \overline{F_C^{r-4-1}} \right\|_2 + \left\| \overline{E(I_{CS})^{r-5-1}} - \overline{F_C^{r-5-1}} \right\|_2 \quad (9)$$

The style loss is defined as:

$$\mathcal{L}_s = \sum_{i=1}^L \left\| \mu(\phi_i(I_{CS})) - \mu(\phi_i(I_s)) \right\|_2 + \left\| \sigma(\phi_i(I_{CS})) - \sigma(\phi_i(I_s)) \right\|_2 \quad (10)$$

where ϕ denotes each layers in the encoder used to calculate the style loss. We give Relu_1_1, Relu_2_1, Relu_3_1, Relu_4_1, Relu_5_1 layers same weight.

According to [12], an identity loss function is introduced to maintain the structure of content image and the style feature of reference image at the same time. The identity loss is defined as:

$$\begin{aligned} \mathcal{L}_{identity} = & \lambda_{identity1} \left(\left\| (I_{CC} - I_C) \right\|_2 + \left\| (I_{SS} - I_S) \right\|_2 \right) \\ & + \lambda_{identity2} \sum_{i=1}^L \left(\left\| \phi_i(I_{CC}) - \phi_i(I_C) \right\|_2 + \left\| \phi_i(I_{SS}) - \phi_i(I_S) \right\|_2 \right) \end{aligned} \quad (11)$$

where I_{CC} and I_{SS} represent the output image synthesized from two same content (or style) images, ϕ_i denotes a layer in the encoder, $\lambda_{identity1}$ and $\lambda_{identity2}$ means identity loss weight.

The $\mathcal{L}_c$, $\mathcal{L}_s$, $\lambda_{identity1}$ and $\lambda_{identity2}$ in the network are the parameter set manually, not by training.

III. EXPERIMENTS

A. Dataset and Experimental Setting

MS-COCO [18] is used as the content image and WikiArt [19] is used as the style image to train the network. Where, there are about 80000 training images from MS-COCO and 10000 training images from WikiArt. In the training process, we first re scale the smaller dimension of the two images to 512 while maintaining the aspect ratio, and then randomly crop a region of 256×256 pixels. Adam optimizer is used in the optimizer with the learning rate of 0.0001, and the batch size is 8 content-style image pairs.

B. Experimental Result Comparison

Qualitative examples. In the training process, we never observe all of the test style images, we randomly selected 6 content pictures and 6 style pictures to show the experimental results. In Fig. 4, we show the sample results generated by this method. In addition, it can be seen that different styles of images can be transferred to different styles according to different styles of input images, and a large domain gap is allowed. The network can transfer an arbitrary style.

Fig. 5 shows the results compared with other models.

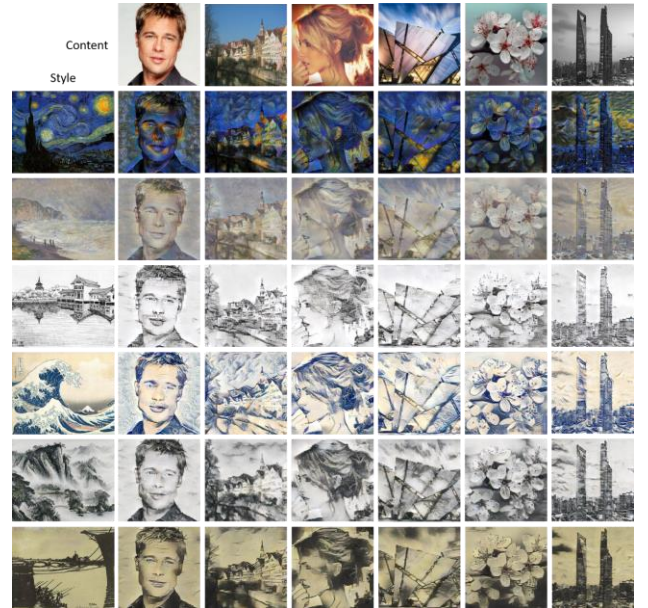


Fig. 4. The multiple style transfer experimental result of dual attention style transfer network

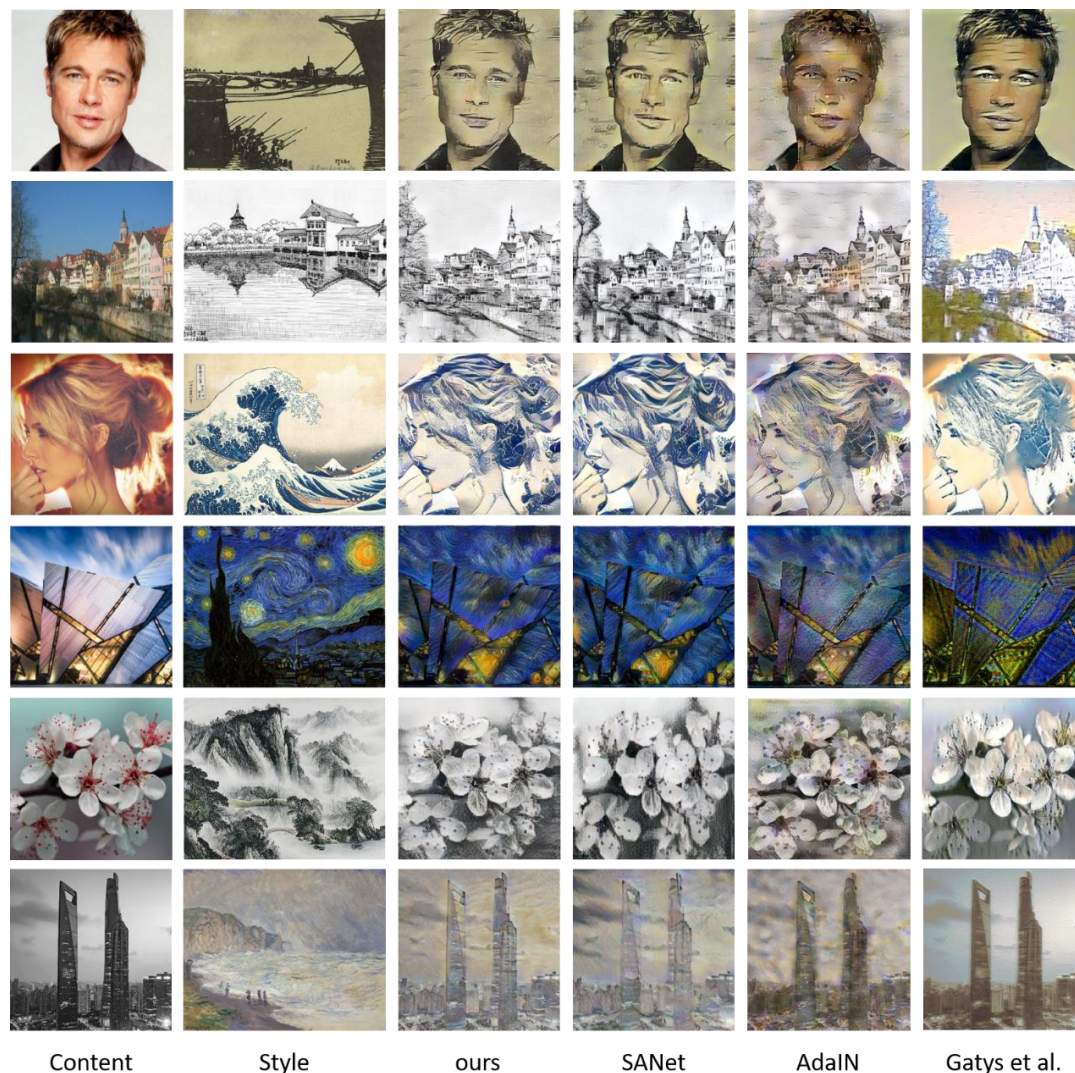


Fig. 5. Comparison between different style transfer methods

The method of Gatys et al. [4] is based on iterative optimization can transfer any style, but the result is not ideal, and the feature extraction of style is not enough. Moreover, for any group of input of content map and style map, using the same parameters may cause destructive impact in some output (as shown in the comparison between the second and fourth lines of Fig. 5). AdaIN [11] synthesizes stylized images by adjusting the mean and variance of content features, which may cause local blur (as shown in the sixth line of Fig. 5), and maintain the color of the original content map to a certain extent, resulting in the overall color of the output image being between the content image and the style image.

Compared with SANet [12], our algorithm can reduce unnecessary features being migrated to the output image, which helps to keep the background clean (as shown in the first line of Fig. 5) and the texture details clear (as shown in the fourth line of Fig. 5). In the processing of detail features, more detailed texture can be captured (as shown in the second, third and fifth lines of Fig. 5).

Finally, it strengthens the grasp of the object boundary in the original image, generates clearer boundary (as shown in the sixth line of Fig. 5) and reduces the distortion when encounter mass of details (as the second line of Fig. 5).



Fig. 6. A more detailed comparison with SANet

A more detailed comparison with SANet is shown in Fig.6. As shown in the first row, style transfer of SANet in the background is not ideal, a large number of strokes at the bottom of the style picture are transferred by mistake, and our method performs well in the background processing. At the same time, compared with SANet, the boundary contour of the bridge and the building in our method is neater and clearer. The same to the second row, our method more accurately transfer the style of the background. At the same time, compared with SANet, our method reduces the distortion of texture and appears smoother. Therefore, in general, our method can obtain higher quality stylized images.

Efficiency. TABLE I. shows the transfer speed comparison of different style transfer methods. The experiment was carried out on a GTX2080Ti device. We have carried out experiments in two dimensions: 256×256 and 512×512 . It can be seen that the algorithm runs at the speed of 66fps in the dimension of 256×256 , and is almost same to the speed of 256×256 in the dimension of 512×512 . Our algorithm is a little slower than SANet, and the speed deviation is at the millisecond level. Therefore, we can consider that our algorithm can make fast style transfer with real-time performance.

TABLE I. COMPARISON OF EFFICIENCY BETWEEN DIFFERENT MODELS

Methods	Time(ms) (256×256)	Time(ms) (512×512)
Gatys et al.	12.841	43.776
AdaIN	0.016	0.025
SANet	0.013	0.014
ours	0.015	0.017

IV. CONCLUSION

This paper proposes a new dual attention style transfer network, which is composed of style attention module and channel attention module. Our approach is more effective. Through the comprehensive attention to features in the two dimensions of style and channel, the decoration of style features can be realized more accurately. The experimental results show that the proposed method can synthesize higher quality images than the previous methods with keeping the applicability and real-time performance. Generally, the method can make arbitrary fast style transfer with higher quality.

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